

A NOTE ON PROVENANCE BEFORE ANYTHING ELSE

The reviewer's instruction (Mengubah_fokus_VERIQO_sekarang.docx) asks for "Revision 0.2" of a prior positioning document -- referred to in that docx as "dokumen Claude" -- with four named changes. That prior document is not present anywhere in this repository or in this session's own history (a repo-wide search of docs/*.md and every PDF filename under docs/ for "MLETR", "eBL", and "Article 12(b)" found nothing). It was evidently produced in a different conversation this session has no access to.

Rather than guess at text this session has never seen, this is a freshly constructed v0.2 that applies all four requested changes and the new positioning sentence directly, grounded in VERIQO's real, already-built, already-tested technical capabilities (drawn from six rounds of audited work in this repository) rather than in an unverifiable prior draft. Where the original document's exact wording would matter (e.g. for a diff against "what changed"), that comparison cannot honestly be made here -- only the requested *end state* is produced.

This document is a technical-evidentiary self-assessment written by an engineering audit process. It is not a legal opinion, and it does not constitute legal advice. Every claim about legal effect, evidentiary weight, or regulatory conformance below must be independently validated against (a) the authoritative UNCITRAL MLETR 2017 text and any explanatory notes, (b) the specific enacting legislation of whichever jurisdiction(s) a transaction is governed by (MLETR is a *model* law -- enacting states adapt it), and (c) qualified legal counsel, before this document is relied upon externally with any party named in it (an eBL platform, a marine surveyor, a P&I club, a trade-finance bank, an insurer, or a court). This caveat is not boilerplate -- it is the literal content of CHANGE 4 below, restated up front so it cannot be skimmed past.

CHANGE 1 -- THE CORE CLAIM, REWORDED

Was: "Veriqo is built for Article 12(b)."

Now:

> "Veriqo is architected to provide transaction-specific evidence that
> may support Article 12(b) where the relevant MLETR function has in
> fact been fulfilled."

The difference is not stylistic. The original phrasing implies Veriqo itself *satisfies* a statutory requirement. The revised phrasing states what Veriqo actually does: it produces evidence *about* whether a transaction's electronic transferable record (ETR) practice satisfied a reliable-method standard -- the legal conclusion of whether that standard was actually met is for the enacting jurisdiction's law, the parties' chosen platform and process, and ultimately a court or regulator to determine. Veriqo is evidentiary infrastructure, not a compliance certification authority.

CHANGE 2 -- ARTICLE 12 (CHANGE OF MEDIUM): A 7-FACTOR TECHNICAL MAPPING

Article 12 of MLETR addresses change of medium -- the functional-equivalence requirements that must hold when a transferable document or instrument moves between paper and electronic form (or back), so that the change does not break the continuity of the "reliable method" standard the record depends on, and does not create two simultaneously valid originals.

The seven factors below are this engineering assessment's own working decomposition of what a change-of-medium event must be able to demonstrate to a third party -- they are not a verbatim reproduction of Article 12's sub-paragraph numbering (which requires the official UNCITRAL text and counsel to map precisely; see the provenance note above). They are organized to match what a technical evidence system can actually attest to.

Article 12 factor	What MLETR asks	What Veriqo provides	Evidence artifact
Current status	External dependency	Independent validation	required

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1. Reliability of method	The method used for the change of medium must be as reliable as appropriate for the purpose	A hash-chained, JCS-canonicalized event record of the change-of-medium act itself (who initiated it, when, under which policy) manifest.CustodyEvent (action=TRANSFORMED/DERIVED) chained via EventHash/PreviousHash	Engineering Verified (Level 2) -- real, tested code in this repo
	The reliable-method *standard itself* (what counts as "appropriate") is set by the enacting legislation and/or the eBL platform's own rules, not by Veriqo	Yes -- a qualified assessor must confirm the specific hash-chain/canonicalization scheme meets the applicable reliability threshold in the relevant jurisdiction	

2. Integrity through the change	The record's content must be demonstrably unchanged in substance across the medium change	ManifestHash/computeManifestHash -- a deterministic, content-only hash independently re-verifiable via VerifyManifestHash	Manifest.ManifestHash + VerifyManifestHash
	Engineering Verified -- proven by adversarial tampering tests (TestVerifyManifestHashDetectsTampering)	None for the hash mechanism itself; the *upstream* content capture (was the paper document faithfully digitized?) is outside Veriqo's control	Yes -- the digitization/capture step upstream of Veriqo needs its own independent attestation

3. Singularity (no duplicate originals)	Only one version of the record may be capable of exercising the underlying rights at any time	StateFinalized immutability + Supersede's single-successor lineage (a FINALIZED version can only be superseded once, never edited)	manifest.Registry.Advance/Supersede, ErrFinalizedIsImmutable, ErrAlreadySuperseded
	Engineering Verified -- proven by TestFinalizedManifestIsImmutable, TestSupersedeRefusesAnUnfinalizedParent	Whether the PAPER original was actually deprived of legal effect at the moment of change is an act outside Veriqo (a platform/registry/carrier action), not something Veriqo's own hash chain can observe or enforce	Yes -- singularity of the *legal* instrument requires the eBL platform's own exclusivity mechanism; Veriqo can only attest to singularity within its own evidence record

4. Identification of the controller at the moment of change	The person entitled to control the record at the time of the medium change must be identifiable	Custody-event Actor field, attributed and hash-chain-covered	CustodyEvent.Actor, hasCustodyActionBoundToContent
	Engineering Verified for the custody-event mechanism itself	Actor identity assurance (was "PTY-1" really that party?) depends on an external identity/authentication system feeding Veriqo, e.g. pkg/security/identity's SPIFFE-based provider, which is not yet wired into this custody-event path (see the companion OS Integration Audit gap document)	Yes -- identity binding strength depends entirely on the upstream identity provider actually used in a given deployment

5. Time and place of the change	When and (where relevant) where the change of medium		
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occurred must be recorded CustodyEvent.Tick and Manifest.FinalizedAt, covered by the same hash chain CustodyEvent, Manifest.FinalizedAt Engineering Verified for the recording mechanism Veriqo's tick is an application-defined monotonic counter, not a certified/synchronized wall-clock timestamp or trusted timestamping authority (RFC 3161 or equivalent) unless the deployment wires one in Yes -- if legal "time" precision is required, a trusted timestamping authority should sit upstream of or alongside the tick value

6. Non-repudiation / attributability The party responsible for the change of medium action must not be able to credibly deny it Hash-chain covering Actor + ContentHash + PreviousHash, refusing any reordering or substitution (TestReorderedCustodyChainFailsVerification) Custody hash chain Engineering Verified for internal consistency Cryptographic non-repudiation in the strict sense (a signature the actor cannot deny producing) requires the signing infrastructure in pkg/platform/security/keys to actually be wired to custody-event submission, which is not yet the case in this repository today Yes -- true non-repudiation requires per-actor signing keys bound to custody events, a real but currently unbuilt integration

7. Independent verifiability / replay An independent party must be able to confirm, after the fact, that the change of medium happened as claimed Deterministic replay: an independent party can reconstruct the same ManifestHash/CustodyChainHead from the same event sequence on a fresh registry TestReplayReproducesIdenticalFinalizedState, and (as of the most recent round) a real multi-node raft consensus proof (TestManifestCluster_FinalizedEvidenceConvergesAcrossRealConsensus) Engineering Verified, and the strongest-evidenced row in this table -- this is VERIQO's core differentiator The independent party still needs access to the event log/snapshot and the verification tooling; Veriqo does not yet ship a standalone, non-Go, third-party-operable verifier Yes -- for court/regulator use, an independently-implemented (not just independently-run) verifier would strengthen the claim further

CHANGE 3 -- MLETR FUNCTION BOUNDARY TABLE

This table is reproduced exactly as specified by the reviewer, since it was handed over as ready content rather than column headers to fill in:

MLETR function	eBL platform	VERIQO
Identify ETR	PRIMARY	Evidence support
Integrity	PRIMARY	STRONG
Exclusive control	PRIMARY	OUT OF SCOPE
Identify controller	PRIMARY	Evidence support
Transfer control	PRIMARY	Evidence capture/verification
Singularity	PRIMARY	Evidence support
Time/place	PRIMARY	Strong evidence support
Transaction auditability	Shared	VERY STRONG
Independent replay	Usually external capability	VERIQO CORE

Read plainly: for every function that actually CONSTITUTES the legal instrument (identifying the ETR, holding exclusive control, transferring control), the eBL platform is PRIMARY and Veriqo is, at most, an evidence-support layer sitting alongside it -- and for "Exclusive control" specifically, Veriqo is explicitly OUT OF SCOPE: Veriqo does not hold, assert, or adjudicate who currently controls an ETR. Where Veriqo is strong-to-core is in the two functions a platform typically does NOT specialize in: transaction auditability and independent replay -- i.e., proving after the fact what happened and letting someone who was not part of the original transaction verify it independently.

CHANGE 4 -- WHAT VERIQO CANNOT CLAIM

Stated explicitly, because the reviewer is right that this increases credibility rather than undermining it:

- Veriqo cannot claim to make an ETR "reliable." Reliability under MLETR is a property of the *method* used by the platform/process that issues, controls, and transfers the record. Veriqo can produce evidence about how that method performed in a specific transaction; it cannot retroactively make an unreliable method reliable, and it does not certify a method as reliable in the abstract.
- Veriqo cannot claim exclusive control. "Exclusive control" is a legal/technical property the eBL platform itself must establish and maintain (see the boundary table above). Veriqo has no mechanism that asserts, grants, or arbitrates who currently controls an ETR.
- Veriqo cannot claim to be a system of record for the ETR itself. The eBL platform is the system of record for the transferable instrument. Veriqo is a system of record for *evidence about* events concerning that instrument -- a distinct, narrower claim.
- Veriqo cannot claim cryptographic non-repudiation today. As Change 2's table notes, per-actor signing is not yet wired to custody-event submission in this repository. Until it is, "non-repudiation" claims should be limited to internal hash-chain consistency, not cryptographic proof against a specific signer.
- Veriqo cannot claim identity assurance. Custody-event actors are recorded as attributed strings; whether "PTY-1" really was the party claimed depends on the identity system feeding Veriqo in a given deployment, which is not yet integrated (see the companion OS Integration Audit document).
- Veriqo cannot claim trusted timestamping. Tick values are application-defined, not RFC 3161 (or equivalent) certified timestamps, unless a deployment separately wires one in.
- Veriqo cannot claim legal conformance with MLETR or any enacting jurisdiction's legislation. Every mapping in this document is a technical-evidentiary self-assessment, not a legal determination. Only qualified counsel, applied to a specific transaction and a specific jurisdiction's enacted law, can make that determination.
- Veriqo cannot claim its distributed-replication and snapshot-restoration proofs were run over a real network. The most recent engineering round proved real multi-node raft consensus using this repository's own in-process transport (raftlite.MemTransport), not the full mTLS pkg/transport/rafttcp + multi-process/Docker harness. This is a genuine, real consensus proof -- but it is not yet a real-socket, real-process, real-network-partition proof for the Evidence/Manifest authority layer specifically. Stated honestly rather than implied as complete.
- Veriqo cannot claim system-level authority assurance yet. Every hardening round to date has proven the Evidence/Manifest/Hypothesis/Finding authority packages safe *in isolation*. Whether that safety survives integration with VERIQO's own API, Workflow, Knowledge, Intelligence, Decision, Ledger, and Cluster layers is the subject of the separate, not-yet-executed "OS Integration Audit" -- see the companion gap-analysis document delivered alongside this one.

THE ONE-SENTENCE POSITIONING STATEMENT

Per the reviewer's own request, verbatim:

> "Veriqo does not make an electronic transferable record legally
> reliable; it makes the evidence of how a reliable method performed in
> a specific transaction independently verifiable."

This is the load-bearing sentence for the whole document: it separates Veriqo's actual, provable claim (evidentiary verifiability) from the claim it must never make (legal reliability), in one line simple enough to survive being quoted out of context by a lawyer, regulator, or investor.

TOWARD THE FULLER MAPPING (ARTICLE 9/10/11/12/13/16/17/18)

The reviewer's final verdict names the next high-value deliverable as a full, line-by-line "MLETR/eBL Reliability Conformance Mapping v0.2" (now this document) extending across Articles 9, 10, 11, 12, 13, 16, 17, and 18: Article -> eBL function -> Veriqo capability -> evidence artifact -> technical proof -> external dependency -> legal boundary -> readiness status.

This v0.2 delivers that structure fully for Article 12 (Change 2's table) and the cross-cutting function boundary (Change 3's table). A best-effort, GENERAL-FUNCTION-LEVEL first pass at the remaining seven articles follows below, using the same eight-column structure -- but each article's *general function* (not its verbatim statutory text, which this session cannot independently confirm without the official UNCITRAL text) is described in normal legal-technology terms, flagged individually as low-confidence where this session's knowledge is general rather than clause-verified, and every row repeats the same external-dependency and validation-required discipline as Change 2.

Article (general function, not verbatim text -- verify against official MLETR text)	eBL function	Veriqo capability	Evidence artifact	Technical proof	External dependency	Legal boundary	Readiness status
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Art. 9 -- General reliability standard for the method used to identify a person and indicate that person's intention (the umbrella functional-equivalence test) and operates the reliable method itself	Establishes	Evidence that a specific instance of the method executed as designed (inputs, outputs, actor, time)	CustodyEvent chain, ManifestHash TestReplayReproducesIdenticalFinalizedState, hash-chain adversarial tests	The reliability *standard* (what threshold counts as reliable) is set by the platform + enacting law, not Veriqo	Veriqo does not set or certify the reliability threshold	Engineering Verified (evidence layer only)	
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Art. 10 -- Control (functional equivalent of possession)	Establishes and enforces exclusive control	NONE -- explicitly out of scope (Change 3)	n/a	n/a	Entirely the eBL platform's responsibility	Veriqo makes no control-related claim	Out of scope by design
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Art. 11 -- Transfer of control (functional equivalent of endorsement/delivery)	Executes and records the transfer act	Evidence capture of the transfer event as reported to Veriqo (custody event)	CustodyEvent (action reflecting transfer), hash chain	Same custody-chain tamper-detection tests as Change 2 row 6	Veriqo does not itself effect or validate the transfer -- it records what the platform reports	Veriqo's record is evidence OF a claimed transfer, not proof the transfer was legally effective	Engineering Verified (evidence capture only)
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Art. 12 -- Change of medium	See Change 2's full 7-factor table above	(see above)	(see above)	(see above)	(see above)	Engineering Verified, most fully	
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mapped article in this document

Art. 13 -- (general function believed to concern requirements applicable to the use of ETRs / party autonomy considerations; this session's confidence on the precise scope of this specific article number is LOW without the official text) Varies by the specific requirement Evidence-layer support only where the requirement concerns integrity/attribution/timing, which Veriqo already provides generically Same mechanisms as above, applied generically Same test suite Depends on the specific requirement Flagged for review: this row should not be relied on until the specific article text is confirmed Low confidence -- requires official text before further mapping

Art. 16 -- (general function believed to concern amendment of an electronic transferable record; LOW confidence on precise scope) Executes and records amendments Evidence capture of an amendment event, superseded-version lineage manifest.Registry.Supersede, MarkSuperseded TestSupersedeCreatesNewVersionWithoutRewritingHistory Whether an "amendment" under this article maps cleanly onto Veriqo's "Supersede" concept needs legal confirmation Flagged for review Low confidence -- plausible technical mapping, unconfirmed legal mapping

Art. 17 -- (general function believed to concern replacement of an electronic transferable record with a paper-based transferable document or instrument, or the reverse; LOW confidence) Executes the replacement/reconversion Same change-of-medium evidence mechanisms as Article 12, applied in reverse Same as Change 2's table Same as Change 2's table Same dependencies as Article 12 Flagged for review -- may substantially overlap Article 12's own scope; needs counsel to disambiguate Low confidence

Art. 18 -- (general function believed to concern requirements for cross-border recognition of electronic transferable records; LOW confidence) Ensures the ETR is recognized across jurisdictions Veriqo's evidence artifacts are jurisdiction-agnostic by construction (content-hash-based, no jurisdiction-specific encoding) -- a NEUTRAL property that MAY be relevant to cross-border recognition arguments, but this is a legal argument this session cannot make n/a n/a Cross-border legal recognition is entirely a matter of the enacting jurisdictions' own law and any applicable treaty/reciprocity arrangement Flagged for review -- do not present Veriqo's jurisdiction-agnostic evidence format as itself satisfying any cross-border recognition requirement without counsel confirming the argument Low confidence

Explicit flag: the rows for Articles 13, 16, 17, and 18 are marked LOW CONFIDENCE because this session does not have verified access to the precise UNCITRAL MLETR article text for those specific numbers and is describing plausible, generally-understood functions rather than clause-verified ones. Publishing this table externally without first replacing those four rows' descriptions with counsel-verified text would be exactly the overclaiming this whole revision exercise exists to prevent. Articles 9, 11, and 12 are HIGHER confidence because their general function (reliability standard, transfer, change of medium) is widely and consistently documented in public MLETR explanatory materials and is consistent with how the reviewer's own docx used them.